



PGDM Batch 2024-26
Trimester 3: Business Analytics
Data Visualization using Power BI(PBAAI32)
Course Project

Submitted To :-

Prof. Mohit Agrawal

Submitted By :-

Devansh Vats

2426/BA/011

Project Overview

- **a. Industry Selected:** "Automobile Industry"
- **b. Domain Focus:** "Sales, Marketing Performance and After-Sales Service & Maintenance"
- **c. Brief Background:**
 - The automotive industry is currently navigating a period of profound transformation driven by several factors, including the shift towards electric vehicles (EVs), advancements in autonomous driving technology, and evolving consumer preferences for digital experiences. In this dynamic environment, understanding and optimizing both **sales performance** and **after-sales service** are paramount for sustained profitability and market leadership.
 - From a **sales and marketing perspective**, competition is fierce, and effective strategies are required to attract new customers, convert leads, and accurately forecast demand. Challenges include optimizing marketing spend, understanding customer demographics, and ensuring dealership performance. Data analytics offers the opportunity to identify top-performing models, assess marketing channel effectiveness, and pinpoint regional sales trends.
 - Concurrently, **after-sales service and maintenance** are not merely cost centers but vital components of the customer lifecycle and significant revenue streams. High-quality service builds customer loyalty, encourages repeat purchases, and maintains brand reputation. Key challenges in this domain include ensuring service efficiency, managing parts inventory, and measuring customer satisfaction. Moreover, with increasingly complex vehicle technologies, particularly EVs, the demands on service centers are evolving rapidly. By integrating the analysis of these two domains, this project aims to provide a holistic view of the automotive business, enabling data-driven decisions that enhance both acquisition and retention strategies.

Problem Statements / Key Questions

This project aims to leverage data analytics to answer specific business questions within the automotive sales, marketing, and after-sales service domains. Addressing these questions will provide actionable insights for improving performance and customer experience.

a. List of Business Questions the Project Aims to Answer:

Sales & Marketing Performance Questions:

1. **Sales Trends & Performance:** What are the monthly, quarterly, and annual sales trends by vehicle model, type (e.g., Sedan, SUV, EV), and region/dealership?
2. **Top/Underperforming Models:** Which vehicle models/types are top sellers, and which are underperforming in terms of units sold and revenue?
3. **Sales Target Achievement:** How does actual sales revenue compare against predefined sales targets at an overall and dealership level?
4. **Average Sale Price:** What is the average sales price per vehicle model/type, and how does it vary across dealerships or regions?
5. **Marketing Channel Effectiveness:** Which marketing channels (e.g., online ads, dealership visits, referrals) are most effective in generating sales?
6. **Customer Demographics (Sales):** What are the key demographic characteristics (e.g., age group, income bracket) of customers purchasing different vehicle models or types?

After-Sales Service & Maintenance Questions:

7. **Service Performance & Efficiency:** What is the average 'First-Time Fix Rate' (FTFR) for service appointments, and how does it vary by service type, vehicle model, or technician?
8. **Common Service Issues:** What are the most common service issues reported by customers, and for which vehicle models are they most prevalent?
9. **Service Cycle Time:** What is the average 'Service Cycle Time' (time from vehicle drop-off to delivery) for different service types, and how can it be optimized?
10. **Service Revenue & Profitability:** What is the total service revenue, and how does it break down by service type (e.g., scheduled maintenance, repairs, parts sales) and dealership?
11. **Customer Satisfaction (Service):** What is the Customer Satisfaction Index (CSI) or Net Promoter Score (NPS) for the service department, and how does it fluctuate over time or across dealerships?
12. **Customer Retention (Service):** What is the customer retention rate for after-sales service, and are there correlations between initial sales experience and subsequent service loyalty?
13. **Parts Management:** Which parts are most frequently replaced or sold during service, and what is their contribution to overall service cost/revenue?

Data Design

The foundation of any robust analytics project is a well-designed data structure. For this project, a star schema approach was adopted, separating data into fact tables (containing transactional data) and dimension tables (containing descriptive attributes). This design facilitates efficient querying and analysis in Power BI.

Data Requirements

To answer the defined business questions, specific data points covering sales transactions, service operations, customer demographics, vehicle specifications, and dealership details are required. Data cleanliness and consistency are paramount for accurate analysis.

Fields (Column Names)

The following tables and their respective columns were designed:

- **Sales Data (Fact Table):**
 - SaleID
 - SaleDate
 - VehicleID
 - CustomerID
 - DealershipID
 - SalePrice
 - DiscountAmount
 - FinancingType
 - MarketingChannel
 - SalespersonID
 - TargetRevenue
- **Service Data (Fact Table):**
 - ServiceOrderID
 - ServiceDate
 - VehicleID
 - CustomerID
 - DealershipID
 - ServiceType
 - ServiceCost
 - PartsCost
 - LaborHours
 - ServiceStatus
 - FirstTimeFix
 - CustomerSatisfactionScore
 - TechnicianID

- **Customer Master (Dimension Table):**
 - CustomerID
 - CustomerName
 - Email
 - Phone
 - Address
 - City
 - State
 - ZipCode
 - AgeGroup
 - IncomeBracket
- **Vehicle Master (Dimension Table):**
 - VehicleID
 - Make
 - Model
 - Year
 - VehicleType
 - EngineType
 - MSRP
- **Dealership Master (Dimension Table):**
 - DealershipID
 - DealershipName
 - Region
 - City
 - State
 - AnnualSalesTarget
 - AnnualServiceTarget

Data Types

The specified data types for each field ensure appropriate storage and calculation:

- **Text:** SaleID, VehicleID, CustomerID, DealershipID, FinancingType, MarketingChannel, SalespersonID, ServiceOrderID, ServiceType, ServiceStatus, FirstTimeFix, TechnicianID, CustomerName, Email, Phone, Address, City, State, ZipCode, AgeGroup, IncomeBracket, Make, Model, VehicleType, EngineType, DealershipName, Region.
- **Number (Whole/Decimal):** SalePrice, DiscountAmount, TargetRevenue, ServiceCost, PartsCost, LaborHours, CustomerSatisfactionScore, Year, MSRP, AnnualSalesTarget, AnnualServiceTarget.
- **Date:** SaleDate, ServiceDate.

Sample Data Format Table

Below are small sample tables illustrating the expected format of the data for each designed table.

Sample Data Format: Sales.csv

A	B	C	D	E	F	G	H	I	J	K
Sale ID	Sale Date	Vehicle ID	Customer ID	Dealership ID	Sale Price	Discount Amount	Financing Type	Marketing Channel	Sales person ID	Target Revenue
S001	2023-01-15	V001	C101	D01	35000	1000	Loan	Online Ad	SP001	36000
S002	2023-01-20	V002	C102	D02	42000	500	Cash	Dealership Visit	SP002	40000
S003	2023-01-22	V003	C103	D01	28000	0	Lease	Referral	SP001	30000

Sample Data Format: Service.csv

A	B	C	D	E	F	G	H	I	J	K	L	M
ServiceOrderID	ServiceDate	VehicleID	CustomerID	DealershipID	ServiceType	ServiceCost	PartsCost	LaborHours	ServiceStatus	FirstTimeFix	CustomerSatisfactionScore	TechnicianID
SRV001	2023-01-25	V001	C101	D01	Scheduled Main	300	100	2.5	Completed	Yes	4	T001
SRV002	2023-02-01	V002	C102	D02	Repair	650	300	4	Completed	No	3	T002
SRV003	2023-02-05	V003	C103	D01	Recall	0	0	1	Completed	Yes	5	T001

Sample Data Format: Customers.csv

A	B	C	D	E	F	G	H	I	J
CustomerID	CustomerName	Email	Phone	Address	City	State	ZipCode	AgeGroup	IncomeBracket
C101	John Doe	john.doe@gmail.com	555-1234	123 Main St	Springfield	IL	62704	41-60	Medium
C102	Jane Smith	jane.s@gmail.com	555-5678	456 Oak Ave	Shelbyville	KY	40065	26-40	High

Sample Data Format: Vehicles.csv

A	B	C	D	E	F	G
VehicleID	Make	Model	Year	Vehicle Type	Engine Type	MSRP
V001	Toyota	Camry	2022	Sedan	Petrol	28000
V002	Honda	CR-V	2023	SUV	Petrol	32000

Sample Data Format: Dealerships.csv

A	B	C	D	E	F	G
DealershipID	DealershipName	Region	City	State	Annual Sales Target	Annual Service Target
D01	AutoWorld Springfield	Midwest	Springfield	IL	50000000	5000000
D02	Premier Motors Shelbyville	South	Shelbyville	KY	45000000	4800000

Data Acquisition

a. Source of Data:

For this project, the data was acquired through **Geek for Geeks, Gemini, Keggle**. Due to the proprietary nature and privacy concerns associated with real-world automotive sales and service data, and the need for a dataset that specifically catered to the defined analytical questions across both sales and service domains, a simulated dataset was generated.

Explanation: The data was created using Microsoft Excel & Google Sheets. This approach allowed for complete control over the dataset's structure, volume, and characteristics, including the intentional introduction of some data quality issues to simulate real-world data challenges and demonstrate robust data transformation capabilities. Approximately [e.g., 8,000] sales records and [e.g., 5,000] service records were generated, spanning a period from [e.g., January 2022 to December 2023], alongside corresponding dimension tables for customers, vehicles, and dealerships.

Data Transformation

Data transformation was a critical phase of the project, ensuring that the raw data was clean, consistent, and correctly formatted for analysis in Power BI. All transformation steps were performed using Power BI's Power Query Editor.

Summary of Cleaning Steps:

The following key data cleaning and transformation steps were applied to each dataset:

1. **Data Loading:** Each of the five raw data files (Sales.csv, Service.csv, Customers.csv, Vehicles.csv, Dealerships.csv) were loaded into Power Query Editor.
2. **Data Type Correction:**
 - Ensured all date columns (SaleDate, ServiceDate) were explicitly set to the 'Date' data type.
 - Verified all numerical fields (e.g., SalePrice, ServiceCost, MSRP, CustomerSatisfactionScore, AnnualSalesTarget) were set to 'Decimal Number' or 'Whole Number' as appropriate.
 - Confirmed all ID fields (e.g., SaleID, CustomerID, VehicleID, DealershipID) were treated as 'Text' to prevent accidental aggregation.
 - All other descriptive fields were set to 'Text'.
3. **Handling Missing Values:**
 - For critical numeric fields such as SalePrice or ServiceCost, any rows containing null values were identified and removed, as these transactions would be incomplete for financial analysis.
 - For optional fields like DiscountAmount or PartsCost, null values were replaced with 0 to ensure numerical integrity without losing records.
 - In the Customer table, missing Email or Phone numbers were left as null or empty strings, as they do not hinder core analytical questions.

4. Handling Incorrect Formats:

- Identified and corrected any cells that contained non-numeric characters in numeric columns (e.g., "N/A" in a price column) by either replacing them with a valid numeric value (e.g., 0) or removing the problematic row if correction was not feasible.
- Ensured all date entries were in a consistent YYYY-MM-DD format to prevent parsing errors.

5. Removing Duplicates:

- Duplicate rows were identified and removed based on the primary key in each table (e.g., SaleID in the Sales table, ServiceOrderID in the Service table, CustomerID in the Customer Master, VehicleID in the Vehicle Master, and DealershipID in the Dealership Master) to ensure data uniqueness and prevent overcounting.

6. Renaming and Standardizing Columns:

- Column names were reviewed and standardized for clarity and consistency (e.g., ensuring no leading/trailing spaces, using PascalCase or similar conventions for easier DAX formula writing).

7. Text Cleaning:

- Applied 'Trim' to all text columns to remove any inadvertent leading or trailing whitespace.
- Used 'Capitalize Each Word' or 'Proper Case' transformations on descriptive text fields (e.g., Make, Model, DealershipName, ServiceType) to ensure consistent presentation in visuals.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the main characteristics of the dataset, identify patterns, detect anomalies, and verify assumptions before in-depth analysis.

a. Data Description

The dataset consists of five interconnected tables, capturing transactional and master data for automotive sales and service operations over a [e.g., two-year] period (from January 2022 to December 2023).

- **Sales Table:** Contains [e.g., 8,500] individual vehicle sales records, detailing transaction date, price, customer, vehicle, dealership, and marketing channel.
- **Service Table:** Comprises [e.g., 5,200] service repair orders, including service date, type, cost, labor hours, and customer satisfaction scores.
- **Customer Master:** Includes [e.g., 980] unique customer profiles with demographic information.
- **Vehicle Master:** Lists [e.g., 115] distinct vehicle models with attributes like make, model, year, and type.
- **Dealership Master:** Details [e.g., 8] individual dealerships, including their location and annual sales/service targets.

Data Statistics and Interpretation

Key statistical observations from the cleaned dataset include:

- **Total Sales Revenue:** The total sales revenue across all dealerships for the period was approximately [e.g., \$350 Million], with an average of [e.g., \$41,176] per vehicle sold. This indicates a focus on [e.g., mid-to-high range] vehicle segments.
- **Total Units Sold:** A total of [e.g., 8,500] vehicles were sold. Sedans and SUVs constituted the majority of sales, but there was a noticeable [e.g., 20% increase] in EV sales year-over-year, hinting at market shifts.
- **Sales Performance Distribution:** Sales activity was highest in [e.g., Q4] of each year, likely due to seasonal promotions. Dealership [e.g., "AutoWorld Springfield"] consistently achieved the highest sales volume, but not always the highest average sale price.
- **Total Service Revenue:** The after-sales service department generated a total revenue of approximately [e.g., \$3.1 Million], with an average service cost per order of [e.g., \$596].
- **Service Order Volume:** Scheduled maintenance accounted for the largest volume of service orders, followed by general repairs. Common repairs included [e.g., brake service, oil changes, tire rotations].
- **First-Time Fix Rate (FTFR):** The overall FTFR across all service orders was [e.g., 85%]. This suggests a good level of efficiency, but areas for improvement exist, especially in complex repairs where FTFR was lower.
- **Customer Satisfaction:** The average Customer Satisfaction Score (CSI) for service was [e.g., 4.2 out of 5]. Dealerships with higher FTFR generally correlated with higher CSI scores.

Data Model Diagram

The logical data model in Power BI was structured as a star schema, which is highly efficient for analytical queries and data visualization. This model comprises two fact tables (Sales and Service) at the center, surrounded by relevant dimension tables (Customers, Vehicles, Dealerships).

Fact and Dimension Tables

- **Fact Tables:**
 - Sales: Contains granular sales transaction data.
 - Service: Contains individual service repair order data.
- **Dimension Tables:**
 - Customers: Provides descriptive attributes about each customer.
 - Vehicles: Contains master data for different vehicle models and types.
 - Dealerships: Offers descriptive information about each dealership location.

Table Relationship Summaries

The relationships were defined as follows:

- **Customers to Sales:**
 - Relationship: CustomerID (Customers) to CustomerID (Sales)
 - Cardinality: One-to-Many (1:*)
 - Cross-filter direction: Single (filters flow from Customers to Sales)
 - Logic: One customer can have multiple sales transactions.
- **Customers to Service:**
 - Relationship: CustomerID (Customers) to CustomerID (Service)
 - Cardinality: One-to-Many (1:*)
 - Cross-filter direction: Single (filters flow from Customers to Service)
 - Logic: One customer can have multiple service orders.
- **Vehicles to Sales:**
 - Relationship: VehicleID (Vehicles) to VehicleID (Sales)
 - Cardinality: One-to-Many (1:*)
 - Cross-filter direction: Single (filters flow from Vehicles to Sales)
 - Logic: One vehicle model can be involved in multiple sales transactions.
- **Vehicles to Service:**
 - Relationship: VehicleID (Vehicles) to VehicleID (Service)
 - Cardinality: One-to-Many (1:*)
 - Cross-filter direction: Single (filters flow from Vehicles to Service)
 - Logic: One vehicle model can be involved in multiple service orders.
- **Dealerships to Sales:**
 - Relationship: DealershipID (Dealerships) to DealershipID (Sales)
 - Cardinality: One-to-Many (1:*)
 - Cross-filter direction: Single (filters flow from Dealerships to Sales)
 - Logic: One dealership can conduct multiple sales transactions.

- **Dealerships to Service:**
 - Relationship: DealershipID (Dealerships) to DealershipID (Service)
 - Cardinality: One-to-Many (1:*)
 - Cross-filter direction: Single (filters flow from Dealerships to Service)
 - Logic: One dealership can manage multiple service orders.

KPI Development (DAX Measures)

Key Performance Indicators (KPIs) and metrics were developed using Data Analysis Expressions (DAX) in Power BI. These measures are dynamic calculations that respond to report filters and selections, providing real-time insights aligned with the business questions.

Sales & Marketing Performance KPIs:

- Total Sales Revenue
- Total Units Sold
- Average Sale Price
- Total Sales Target
- Sales Target Achievement %
- Sales Conversion Rate (simplified for channel contribution)

After-Sales Service & Maintenance KPIs:

- Total Service Revenue
- Total Service Orders
- Average Service Cost per Order
- First Time Fix Count
- First Time Fix Rate %
- Average CSI (Customer Satisfaction Index)
- Total Service Target
- Service Target Achievement %

Here are the DAX measures developed, along with their purpose and the corresponding code:

Sales & Marketing Performance KPIs:

1. Total Sales Revenue

- **Purpose:** Measures the total monetary value generated from all vehicle sales. This is a fundamental metric for assessing overall sales performance.
- **DAX Code:**

```
Total Sales Revenue = SUM(Sales[SalePrice])
```

2. Total Units Sold

- **Purpose:** Counts the total number of individual vehicles sold. Essential for understanding sales volume and popularity of models.
- **DAX Code:**

```
Total Units Sold = COUNTROWS(Sales)
```

3. Average Sale Price

- **Purpose:** Calculates the average price at which vehicles are sold. Useful for analyzing pricing strategies and product mix profitability.
- **DAX Code:**

```
Average Sale Price = DIVIDE([Total Sales Revenue], [Total Units Sold], 0)
```

4. Total Sales Target

- **Purpose:** Aggregates the annual sales targets from all dealerships. Serves as a baseline for measuring performance against goals.
- **DAX Code:**

```
Total Sales Target = SUM(Dealerships[AnnualSalesTarget])
```

5. Sales Target Achievement %

- **Purpose:** Shows the percentage of the sales target that has been achieved. Crucial for performance tracking and identifying areas of over or under-performance.
- **DAX Code:**

```
Sales Target Achievement % = DIVIDE([Total Sales Revenue], [Total Sales Target], 0)
```

6. Sales Conversion Rate (by Marketing Channel)

- **Purpose:** (Simplified for this project to show contribution per channel) This measure calculates the proportion of total sales contributed by a specific marketing channel. *Note: A true conversion rate would require a 'Leads' table.*
- **DAX Code:** (This would often be shown in a visual, not a single measure. Here's a measure to help calculate sales contribution if you group by channel).

```
Sales % by Channel =  
DIVIDE([Total Units Sold],  
    CALCULATE([Total Units Sold], ALL(Sales[MarketingChannel])))
```

After-Sales Service & Maintenance KPIs:

7. Total Service Revenue

- **Purpose:** Calculates the total income generated from all service and maintenance operations. Key financial metric for the service department.
- **DAX Code:**

```
Total Service Revenue = SUM(Service[ServiceCost])
```

8. Total Service Orders

- **Purpose:** Counts the total number of service repair orders. Indicates the volume of work handled by the service department.

```
DAX Code: Total Service Orders = COUNTROWS(Service)
```

9. Average Service Cost per Order

- **Purpose:** Determines the average cost incurred per service appointment. Helps in analyzing efficiency and pricing of services.
- **DAX Code:**

```
Average Service Cost per Order = DIVIDE([Total Service Revenue],  
[Total Service Orders], 0)
```

10. First Time Fix Count

- **Purpose:** Counts the number of service orders that were resolved successfully on the first visit. Used as a component for FTFR.
- **DAX Code:**

```
First Time Fix Count = CALCULATE(COUNTROWS(Service),  
Service[FirstTimeFix] = "Yes")
```

11. First Time Fix Rate %

- **Purpose:** Measures the percentage of service issues resolved during the initial appointment. A crucial indicator of service quality and efficiency.
- **DAX Code:**

```
First Time Fix Rate % = DIVIDE([First Time Fix Count], [Total  
Service Orders], 0)
```

12. Average CSI (Customer Satisfaction Index)

- **Purpose:** Calculates the average customer satisfaction score from service feedback. Directly reflects customer experience.
- **DAX Code:**

```
Average CSI = AVERAGE(Service[CustomerSatisfactionScore])
```

13. Total Service Target

- **Purpose:** Aggregates the annual service revenue targets for all dealerships. For measuring the service department's financial goal attainment.
- **DAX Code:**

```
Total Service Target = SUM(Dealerships[AnnualServiceTarget])
```

14. Service Target Achievement %

- **Purpose:** Shows the percentage of the service revenue target that has been achieved. Similar to sales target, but for the service side of the business.
- **DAX Code:**

```
Service Target Achievement % = DIVIDE([Total Service Revenue],  
[Total Service Target], 0)
```